Roll No. | | | | | | | |

Mid Semester Examination March 2025

Name of the Program: B.Tech (CE) *Semester:* *VIII*

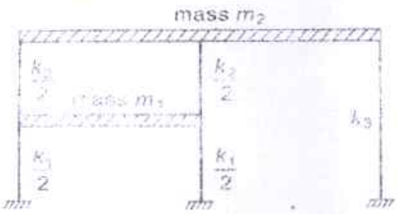
Name of the Course: Earthquake resistant Design of buildings *Course Code:* TCE 801

Time: 90 minutes Maximum Marks: 50

Note:

(i) Answer all the **questions** by choosing ***any one of the sub questions.***

(ii) Each question carries 10 marks

Q1	(10 marks)	CO1
(a)	What is the measure of magnitude? What is the basic principle behind Richter magnitude?	
	OR	
(b)	Explain the Elastic rebound theory. Also briefly describe the continental drift theory put forth by Alfred Wegener.	
Q2	(10 marks)	CO2
(a)	Distinguish between Rayleigh waves and love waves.	
	OR	
(b)	For the system shown below formulate the equation of motion governing the undamped free vibration.  	
Q3	(10 marks)	CO2
(a)	What is meant by undercritical, critical and overcritical damping? Give mathematical expressions.	
	OR	
(b)	Explain the following terms: i. Single degree of freedom system ii. Free vibration iii. Forced vibration iv. Time period v. Damping	
Q4	(10 marks)	CO1&CO2
(a)	Derive the solution $x(t)$ for a damped free vibration case if $\zeta < 1$.	
	OR	
(b)	For a damped system, *m, c, and k* are known to be m = 1 kg, c = 2 kg/s, k = 10 N/m. Calculate the value of ζ and ω_n. Is the system overdamped, underdamped, or critically damped?	

Q5	(10 marks)	
(a)	Explain Modified Mercalli intensity (MMI) scale in detail.	CO1&CO2
	OR	
(b)	An empty elevated water tank is pulled by a steel cable by applying a 30kN force. The tank is pulled horizontally by 5cm. The cable is suddenly cut, and the resulting free vibration is recorded. At the end of five complete cycles, the time is 2s and the amplitude is 2cm. Determine the damping ratio, natural period of undamped vibration and effective stiffness	